

10. Določiti D_f , izračunaj

$$F(y) = \int_{-\infty}^{\infty} \underbrace{\frac{y}{1+x^2y^2}}_{f(x,y)} dx = \int_{-\infty}^0 f(x,y) dx + \int_0^{\infty} f(x,y) dx \stackrel{f(y) \text{ lična}}{=} 2 \int_0^{\infty} f(x,y) dx$$

$F(y)$ konv., ker $\forall \infty$ pada $\sim \frac{1}{x^2}$ za $\forall y \Rightarrow D_f = \mathbb{R}$

$$F(y) = 2y \int_0^{\infty} \frac{1}{1+(xy)^2} dx \stackrel{\substack{x|y|=u \\ y \neq 0 \\ y dx = du}}{=} 2 \frac{y}{|y|} \arctg u = \pi \operatorname{sgn} y$$

$y=0$: $F(0)=0$ $\rightarrow$ je skladno z $\boxed{F(y) = \pi \cdot \operatorname{sgn}(y)}$

Na katerih intervalih $J \subseteq [0, \infty)$ $G(y) = \int_0^{\infty} \frac{y}{1+x^2y^2} dx$ konv, enakomerno?

Domneva: konv. je en. na (a, b) ; $0 < a < b \leq \infty$

$$\int_c^{\infty} \frac{y}{1+x^2y^2} dx \stackrel{y \neq 0}{=} \arctg xy \Big|_c^{\infty} \\ \rightarrow = \begin{cases} \frac{\pi}{2} - \arctg(cy); & y \neq 0 \\ 0; & y = 0 \end{cases}$$

1) $J = (a, b)$, $a > 0$:

Vzamemo $\varepsilon > 0$

$$\left| \int_1^{\infty} \frac{y}{1+x^2y^2} dx \right| = \left| \frac{\pi}{2} - \arctg cy \right| = \frac{\pi}{2} - \arctg cy < \frac{\pi}{2} - \arctg ca$$

$\rightarrow$ vemo, da je $\lim_{x \rightarrow \infty} \arctg x = \frac{\pi}{2} \Rightarrow \exists A < \infty: \forall x \geq A: \left| \frac{\pi}{2} - \arctg x \right| < \varepsilon$

$\rightarrow$ če $ca \geq A \Rightarrow c \geq \frac{A}{a} \rightarrow \checkmark$

ocenimo navzdol,
da je razlika
max. možna

Pokažimo že, da konv. ni enak. na $J = (0, b)$

Naj bo $\varepsilon = \frac{\pi}{4}$, c poljubno. če vzamemo $y = \frac{1}{c} \Rightarrow \frac{\pi}{2} - \arctg(c \cdot \frac{1}{c}) = \frac{\pi}{4} = \varepsilon$
 $\Rightarrow \Rightarrow \varepsilon$ in ne $< \varepsilon$.

(13.) Pokaži, da je $F(y) = \int_0^{\pi/2} \ln(y^2 - \sin^2 \varphi) d\varphi$ zvezen pri $y=1$.
 $y \geq 1$

za $y > 1$: $F(y) = \pi \ln \frac{y + \sqrt{y^2 - 1}}{2}$

$y=1$: $F(1) = \int_0^{\pi/2} \ln(1 - \sin^2 \varphi) d\varphi = \int_0^{\pi/2} \ln(\cos^2 \varphi) d\varphi = 2 \int_0^{\pi/2} \ln(\cos \varphi) d\varphi$
 $\uparrow$ problem pri $\pi/2$

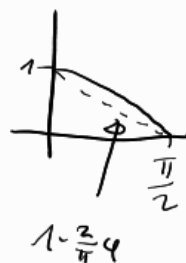
$\rightarrow$ preveriti je treba enak. konv. na $y \in [1, 2]$
 Weierstrassov krit.

$$\ln(\cos^2 \varphi) \leq \ln(y^2 - \sin^2 \varphi) \leq \ln y^2 \leq \ln 4$$

$$\Rightarrow |\ln(y^2 - \sin^2 \varphi)| \leq \ln 4 + 2 |\ln \cos \varphi|$$

ali je integrabilna na $[0, \pi/2]$?

$$\begin{aligned} \int_0^{\pi/2} (\ln 4 + 2 |\ln \cos \varphi|) d\varphi &\leq \pi \ln 2 - 2 \int_0^{\pi/2} \ln(\cos \varphi) d\varphi \leq \\ &\leq \pi \ln 2 - 2 \int_0^{\pi/2} \ln(1 - \frac{2}{\pi} \varphi) d\varphi = \\ &= \pi \ln 2 + \pi \int_1^0 \ln t dt = \pi \ln 2 - \pi (t \ln t - t) \Big|_1^0 = \\ &\quad 1 - \frac{2}{\pi} \varphi = t \quad \Rightarrow \pi \ln 2 + \pi \\ &\quad -\frac{2}{\pi} d\varphi = dt \end{aligned}$$



$\Rightarrow$ funkc. je integrabilna $\stackrel{\text{Weierstrass}}{\Rightarrow} F(y)$ konv. enak.

$$\Rightarrow F(1) = \pi \lim_{y \rightarrow 1} \ln \frac{y + \sqrt{y^2 - 1}}{2} = \pi \ln \frac{1}{2} = -\pi \ln 2$$

Odvajanje pospl. int.

- 1) pospl. integral $\exists$ za nek $y \in J$
- 2) za $\forall x \in (a, b)$ in $\forall y \in J$ je $f(x, y)$ zvezna v (x, y)
- 3) za $\forall x \in (a, b)$ in $\forall y$ iz notranjosti J $\frac{\partial f(x, y)}{\partial y} \exists$ in je zvezen
- 4) pospl. integral $\int_a^b \frac{\partial f(x, y)}{\partial y} dx$ enakomerno konv. za $y \in J$.

$\Rightarrow F(y)$ je zvezno odv. in

$$\underline{F'(y) = \int_a^b \frac{\partial f(x, y)}{\partial y} dx}$$